MUSICAL INSTRUMENT
E-COMMERCE MANAGEMENT SYSTEM

DATABASE RELATIONSHIP ANALYSIS USING JOINS

Chapter 9

1. Combining Multiple Tables

Combining multiple tables means retrieving related information from two or more database tables using SQL JOIN operations. In the Musical Instrument E-Commerce Management System, joins can be used to combine customer, order, product, and payment details for order tracking and reporting.

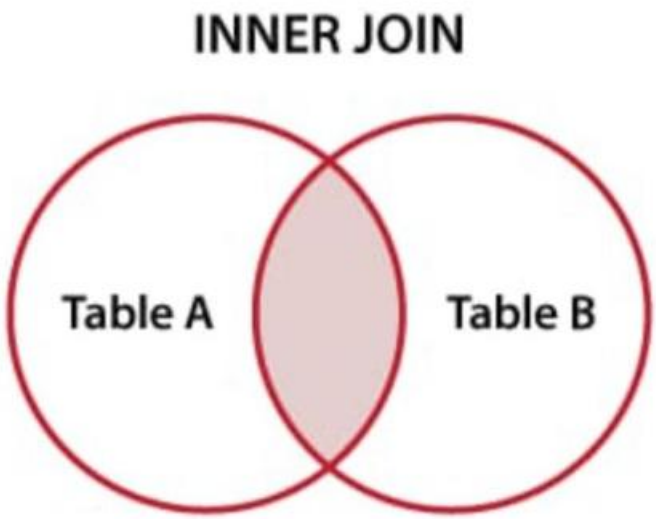
Tables and relationships used in this chapter:

Table	Important columns	Relationship
Customer	CUSTOMERID, CUSTOMERNAME, AGE, CITY, ADDRESS	Customer.CUSTOMERID → Orders.CUSTOMER_ID
Orders	ORDER_ID, CUSTOMER_ID, ORDER_DATE	Orders.ORDER_ID → Payment.ORDER_ID
Payment	PAYMENT_ID, PAYMENT_METHOD, PAYMENT_STATUS, PAYMENT_DATE, AMOUNT, ORDER_ID	Payment.ORDER_ID references Orders.ORDER_ID
Product	Product_ID, Product_Name, Brand, Category, Price, Stock	Product is linked to orders through an order-detail table

The Customer, Orders, and Payment column names above are based on the table structures provided. The Product columns follow the project design supplied. Product reports require an order-detail table (for example, Order_Details) containing ORDER_ID, PRODUCT_ID, and QUANTITY.

2. INNER JOIN

INNER JOIN returns only the rows that have matching values in the joined tables.



2.1 Customer and Orders

Aim: Display customers who have matching orders.

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.First_Name,
  4      C.Last_Name,
  5      O.Order_ID,
  6      O.Order_Date,
  7      O.Total_Amount
  8  FROM Customer C
  9  INNER JOIN Orders O
 10      ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
         101 Rahul
Sharma                                                401 01-JUL-26
         2999

         102 Priya
Kumar                                                 402 03-JUL-26
         1999

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
```

2.2 Orders and payment

```
SQL> SELECT
  2      O.ORDER_ID,
  3      P.PAYMENT_ID,
  4      P.PAYMENT_MODE,
  5      P.PAYMENT_DATE,
  6      P.PAYMENT_AMOUNT,
  7      P.PAYMENT_STATUS
  8  FROM ORDERS O
  9  INNER JOIN PAYMENT P
 10      ON O.ORDER_ID = P.ORDER_ID;

  ORDER_ID PAYMENT_ID PAYMENT_MODE         PAYMENT_D PAYMENT_AMOUNT
---------- ---------- -------------------- --------- --------------
PAYMENT_STATUS
--------------------
       101        501 UPI                  29-SEP-26           1898
Successful

       102        502 Credit Card          29-SEP-26           1299
Successful
```

2.3 Product and Order details

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.First_Name,
  4      C.Last_Name,
  5      O.Order_ID,
  6      O.Order_Date,
  7      O.Total_Amount
  8  FROM Customer C
  9  INNER JOIN Orders O
 10      ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
        101 Rahul
Sharma                                                401 01-JUL-26
        2999

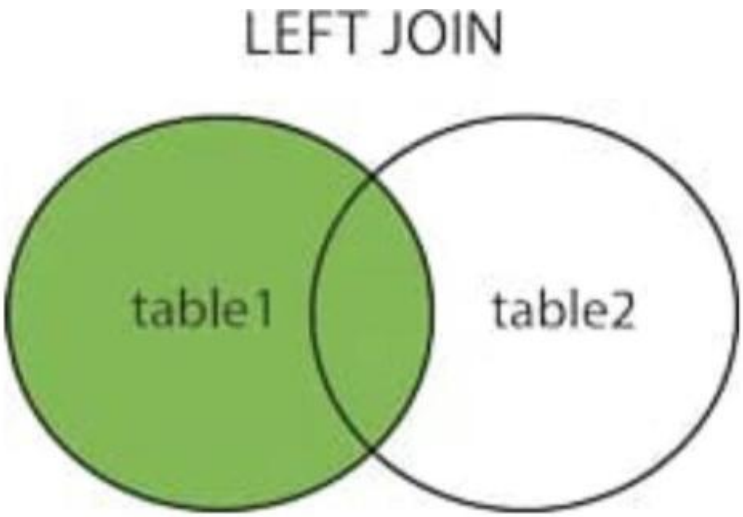
        102 Priya
Kumar                                                 402 03-JUL-26
        1999

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------

        103 Arun
Raj                                                   403 05-JUL-26
        4999
```

3.LEFT JOIN LEFT JOIN

displays all records from the left table and matching records from the right table.



3.1 Display all customers and their orders

```
SQL> SELECT
  2       C.CUSTOMER_ID,
  3       C.CUSTOMER_NAME,
  4       O.ORDER_ID,
  5       O.ORDER_DATE,
  6       O.TOTAL_AMOUNT
  7  FROM CUSTOMERS C
  8  RIGHT JOIN ORDERS O
  9       ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
----------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- --------- ------------


       401 01-JUL-26          2999


       402 03-JUL-26          1999

CUSTOMER_ID
-----------
CUSTOMER_NAME
----------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- --------- ------------


       403 05-JUL-26          4999
```

4.RIGHT JOIN

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.CUSTOMER_NAME,
  4      O.ORDER_ID,
  5      O.ORDER_DATE,
  6      O.TOTAL_AMOUNT
  7  FROM CUSTOMERS C
  8  RIGHT JOIN ORDERS O
  9      ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
----------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- --------- ------------


       401 01-JUL-26         2999



       402 03-JUL-26         1999

CUSTOMER_ID
-----------
CUSTOMER_NAME
```

4.1 Display all orders and customer details

```
SQL> SELECT
  2      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3      O.ORDER_ID,
  4      O.ORDER_DATE,
  5      P.PRODUCT_NAME,
  6      OD.QUANTITY,
  7      OD.UNIT_PRICE,
  8      PY.PAYMENT_MODE,
  9      PY.PAYMENT_STATUS
 10  FROM CUSTOMER C
 11  INNER JOIN ORDERS O
 12      ON C.CUSTOMER_ID = O.CUSTOMER_ID
 13  INNER JOIN ORDER_DETAILS OD
 14      ON O.ORDER_ID = OD.ORDER_ID
 15  INNER JOIN PRODUCT P
 16      ON OD.PRODUCT_ID = P.PRODUCT_ID
 17  LEFT JOIN PAYMENT PY
 18      ON O.ORDER_ID = PY.ORDER_ID;

CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT
---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE PAYMENT_MODE          PAYMENT_STATUS
---------- ---------- -------------------- --------------------
Arun Raj
       403 05-JUL-26
Gaming Keyboard
         1       2799

CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT
---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE PAYMENT_MODE          PAYMENT_STATUS
```

5. Complete Order details

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      O.ORDER_ID,
  5      O.ORDER_DATE,
  6      O.ORDER_STATUS,
  7      P.PRODUCT_ID,
  8      P.PRODUCT_NAME,
  9      OD.QUANTITY,
 10      OD.UNIT_PRICE,
 11      OD.SUBTOTAL,
 12      O.TOTAL_AMOUNT,
 13      PY.PAYMENT_MODE,
 14      PY.PAYMENT_STATUS
 15  FROM CUSTOMER C
 16  INNER JOIN ORDERS O
 17      ON C.CUSTOMER_ID = O.CUSTOMER_ID
 18  INNER JOIN ORDER_DETAILS OD
 19      ON O.ORDER_ID = OD.ORDER_ID
 20  INNER JOIN PRODUCT P
 21      ON OD.PRODUCT_ID = P.PRODUCT_ID
 22  LEFT JOIN PAYMENT PY
 23      ON O.ORDER_ID = PY.ORDER_ID
 24  ORDER BY O.ORDER_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID ORDER_DAT ORDER_STATUS                   PRODUCT_ID
---------- --------- ------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL TOTAL_AMOUNT PAYMENT_MODE
---------- ---------- ---------- ------------ ----------------
PAYMENT_STATUS
--------------------
        101

CUSTOMER_ID
```

6.Customer purchase history

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      O.ORDER_ID,
  5      O.ORDER_DATE,
  6      O.ORDER_STATUS,
  7      P.PRODUCT_NAME,
  8      OD.QUANTITY,
  9      OD.UNIT_PRICE,
 10      OD.SUBTOTAL,
 11      O.TOTAL_AMOUNT
 12  FROM CUSTOMER C
 13  INNER JOIN ORDERS O
 14      ON C.CUSTOMER_ID = O.CUSTOMER_ID
 15  INNER JOIN ORDER_DETAILS OD
 16      ON O.ORDER_ID = OD.ORDER_ID
 17  INNER JOIN PRODUCT P
 18      ON OD.PRODUCT_ID = P.PRODUCT_ID
 19  ORDER BY C.CUSTOMER_ID, O.ORDER_DATE;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID ORDER_DAT ORDER_STATUS
---------- --------- ------------------------------
PRODUCT_NAME
--------------------------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL TOTAL_AMOUNT
---------- ---------- ---------- ------------
         101
Rahul Sharma
       401 01-JUL-26 Delivered

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID ORDER_DAT ORDER_STATUS
```

7.Customer purchase history with payment

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      O.ORDER_ID,
  5      O.ORDER_DATE,
  6      O.ORDER_STATUS,
  7      P.PRODUCT_NAME,
  8      OD.QUANTITY,
  9      OD.UNIT_PRICE,
 10      OD.SUBTOTAL,
 11      O.TOTAL_AMOUNT,
 12      PY.PAYMENT_MODE,
 13      PY.PAYMENT_AMOUNT,
 14      PY.PAYMENT_STATUS
 15  FROM CUSTOMER C
 16  INNER JOIN ORDERS O
 17      ON C.CUSTOMER_ID = O.CUSTOMER_ID
 18  INNER JOIN ORDER_DETAILS OD
 19      ON O.ORDER_ID = OD.ORDER_ID
 20  INNER JOIN PRODUCT P
 21      ON OD.PRODUCT_ID = P.PRODUCT_ID
 22  LEFT JOIN PAYMENT PY
 23      ON O.ORDER_ID = PY.ORDER_ID
 24  ORDER BY C.CUSTOMER_ID, O.ORDER_DATE;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID ORDER_DAT ORDER_STATUS
---------- --------- ------------------------------
PRODUCT_NAME
--------------------------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL TOTAL_AMOUNT PAYMENT_MODE
---------- ---------- ---------- ------------ ----------
PAYMENT_AMOUNT PAYMENT_STATUS
-------------- --------------------
        101
```

8.Multi-Table business report

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME
  4      COUNT(DISTINCT O.ORDER_ID) AS TOTAL_ORDERS,
  5      SUM(OD.SUBTOTAL) AS TOTAL_PURCHASE,
  6      SUM(OD.QUANTITY) AS TOTAL_PRODUCTS
  7  FROM CUSTOMER C
  8  INNER JOIN ORDERS O
  9      ON C.CUSTOMER_ID = O.CUSTOMER_ID
 10  INNER JOIN ORDER_DETAILS OD
 11      ON O.ORDER_ID = OD.ORDER_ID
 12  GROUP BY
 13      C.CUSTOMER_ID,
 14      C.FIRST_NAME,
 15      C.LAST_NAME
 16  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
------------------------------------------------------------
TOTAL_ORDERS TOTAL_PURCHASE TOTAL_PRODUCTS
------------ -------------- --------------
        104
Sneha Reddy
           1           5998              2

        101
Rahul Sharma
           1           5997              3
```

9.Payment transaction report

```
SQL> SELECT
  2      PY.PAYMENT_ID,
  3      O.ORDER_ID,
  4      C.CUSTOMER_ID,
  5      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  6      PY.PAYMENT_MODE,
  7      PY.PAYMENT_DATE,
  8      PY.PAYMENT_AMOUNT,
  9      PY.PAYMENT_STATUS,
 10      O.TOTAL_AMOUNT,
 11      O.ORDER_STATUS
 12  FROM PAYMENT PY
 13  INNER JOIN ORDERS O
 14      ON PY.ORDER_ID = O.ORDER_ID
 15  INNER JOIN CUSTOMER C
 16      ON O.CUSTOMER_ID = C.CUSTOMER_ID
 17  ORDER BY PY.PAYMENT_DATE DESC;

PAYMENT_ID   ORDER_ID CUSTOMER_ID
---------- ---------- -----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
PAYMENT_MODE         PAYMENT_D PAYMENT_AMOUNT PAYMENT_STATUS       TOTAL_AMOUNT
-------------------- --------- -------------- -------------------- ------------
ORDER_STATUS
------------------------------
       501        401         101
Rahul Sharma
UPI                  29-OCT-26           2999 Completed                    2999
Delivered


PAYMENT_ID   ORDER_ID CUSTOMER_ID
---------- ---------- -----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
PAYMENT_MODE         PAYMENT_D PAYMENT_AMOUNT PAYMENT_STATUS       TOTAL_AMOUNT
-------------------- --------- -------------- -------------------- ------------
ORDER_STATUS
------------------------------
       502        402         102
Priya Kumar
Card                 29-OCT-26           1999 Completed                    1999
Shipped
```